

ORIGINAL
ARTICLE

Miocene and Pliocene colonization of the Central American Isthmus by the weakly electric fish *Brachyhypopomus occidentalis* (Hypopomidae, Gymnotiformes)

Sophie Picq^{1,2*}, Fernando Alda¹, Rüdiger Krahe² and Eldredge Bermingham^{1,2}

¹Smithsonian Tropical Research Institute, Balboa, Ancón, Panama, ²Biology Department, McGill University, Montreal, Quebec H3A 1B1, Canada

ABSTRACT

Aim We present a molecular phylogenetic and biogeographical analysis of *Brachyhypopomus occidentalis*, one of the few gymnotiform electric fish in Central America, to further understand the colonization and diversification processes of primary freshwater fishes over the Central American Isthmus.

Location Lower Central America.

Methods We used Bayesian and maximum-likelihood phylogenetic reconstructions using mitochondrial [cytochrome *c* oxidase subunit I (*COI*) and ATP synthase 6 and 8 (*ATPase 8/6*)] and nuclear (*RAG1* and rhodopsin) genes and extensive geographical sampling, together with molecular clock analyses and tests of biogeographical scenarios to assess the timing and mode of dispersal and diversification.

Results We identified high levels of phylogeographical structure, with a highly divergent lineage composed of individuals from western Atlantic Panama (Bocas), sister to all trans-Andean South American and Central American lineages. The Pacific slope of Panama showed surprisingly little genetic structure compared with the Atlantic slope. Molecular-clock and biogeographical analyses support two waves of colonization originating from South America: a first dispersal event in the late Miocene with the Bocas lineage as the only relict, and a second major colonization in the late Pliocene leading to the establishment of *B. occidentalis* in all central and eastern Panama drainages.

Main conclusions The genetic structure of *B. occidentalis* over the Isthmian landscape reflects the progressive, complex and dynamic geological evolution of the region. Our results support multiple colonization events, with an ancient Miocene dispersal event followed by a recent rapid expansion in the late Pliocene, probably facilitated by the final closure of the Isthmus, which provided an important corridor.

Keywords

Central America, diversification, electric fishes, freshwater fishes, historical biogeography, ichthyological provinces, Isthmus of Panama, knifefishes, molecular clock, multiple colonization.

*Correspondence: Sophie Picq, Biology Department, McGill University, 1205, Docteur Penfield, Montreal, QC H3A 1B1, Canada. E-mail: sophiepicq@gmail.com

INTRODUCTION

The complex and dynamic palaeogeography of lower Central America (LCA), which originated as a volcanic island arc in the Late Cretaceous and became a land bridge connecting South and Central America in the late Pliocene, has greatly

contributed to shaping the occurrence and distribution of Central American fauna (Stehli & Webb, 1985). The final closure of the Isthmus of Panama has traditionally been associated with major climatic (Lear *et al.*, 2003), ocean-circulation (Burton *et al.*, 1997) and biotic changes, such as the onset of the Great American Interchange (Stehli & Webb, 1985;

Marshall, 1988). Nonetheless, the exact timing of the final closure, estimated at 3–3.5 million years ago (Ma) (Coates *et al.*, 1992, 2004; Coates & Obando, 1996), remains a controversial issue, as recent geological work suggests an early Miocene initiation of the collision of the Central American arc with South America (Farris *et al.*, 2011), and the possibility of a total closure of the Isthmus 15 Ma (Montes *et al.*, 2012a,b). Regardless of the precise age of the land bridge, this complex terrestrial corridor offers an exceptional opportunity to study dispersal, community assembly and diversification.

Molecular tools have played an important role in understanding the evolutionary history and the timing of faunal dispersal across South America (SA) and Central America (CA), particularly for taxa with a poor fossil record, such as frogs (Weigt *et al.*, 2005; Pinto-Sánchez *et al.*, 2012) and birds (Weir *et al.*, 2009). Furthermore, special attention has been devoted to the mode, timing and number of colonizations of CA by freshwater fishes (Bermingham & Martin, 1998; Perdices *et al.*, 2002, 2005; Reeves & Bermingham, 2006; Ornelas-García *et al.*, 2008; Lovejoy *et al.*, 2010; Alda *et al.*, 2013b; Říčan *et al.*, 2013). Interestingly, the freshwater fish diversity of CA is significantly different from that of SA, although it is mostly of SA origin (Myers, 1966; Bussing, 1976, 1985): primary freshwater fishes (mainly Characiformes, Siluriformes and Gymnotiformes) dominate South American waters, but they are outnumbered by secondary freshwater fishes in CA (mostly Perciformes and Cyprinodontiformes). In the location of our study, however, corresponding to the southernmost part of CA, the freshwater fish fauna is closer to that of SA but less diverse.

According to Myers (1966) and Bussing (1976, 1985), who were among the first to generate dispersal scenarios to explain the composition of the CA ichthyofauna, the scarcity of primary freshwater fishes could be attributed to the recent age of LCA and to the fishes' low dispersal ability, as they are presumed to be physiologically intolerant to seawater. Subsequently, multiple, non-mutually-exclusive hypotheses have been proposed to explain potential colonization events from the Cretaceous to the Pliocene (Bussing, 1976, 1985; Lundberg *et al.*, 1998; Iturralde-Vinent & MacPhee, 1999; Perdices *et al.*, 2002; Smith & Bermingham, 2005; Hrbek *et al.*, 2007; Říčan *et al.*, 2013). The late Pliocene emergence of the Isthmus of Panama has nonetheless been traditionally considered the main trigger for the colonization and dispersal of freshwater fishes into CA (Bermingham & Martin, 1998; Martin & Bermingham, 2000; Perdices *et al.*, 2002; Reeves & Bermingham, 2006). However, some studies allow for isolated dispersal events dating back to the early Pliocene or late Miocene (e.g. *Rhamdia*, 6.5–5.6 Ma; Perdices *et al.*, 2002; *Brachyhypopomus*, late Miocene; Bermingham & Martin, 1998), whereas others suggest that large-scale colonization occurred before the Pliocene (e.g. *Gymnotus*, 15 Ma, Lovejoy *et al.*, 2010; *Astyanax*, 8 Ma, Ornelas-García *et al.*, 2008). Earlier colonization would imply either dispersal through marine environments or an older age for the rise of the Isthmus (Farris *et al.*, 2011; Montes *et al.*, 2012a,b).

The distribution and assemblage of species along the LCA landscape has allowed several biogeographical provinces to be defined (Miller, 1966; Bussing, 1976; Smith & Bermingham, 2005). The most recent and complete study identifies five biogeographical provinces in Panama (Bocas, Chiriquí, Santa María, Chagres and Tuira; Smith & Bermingham, 2005; Fig. 1), which represent regions with a shared history of colonization and diversification of species. Phylogeographical relationships between these regions can inform us on the colonization process of LCA.

Among Gymnotiformes, five genera have successfully colonized CA (*Gymnotus*, *Eigenmannia*, *Sternopygus*, *Apteronotus* and *Brachyhypopomus*). In Panama, all genera occur in the eastern drainages of Tuira and Bayano – the biogeographical province of Tuira. Some, such as *Eigenmannia* and *Apteronotus*, are restricted to this region (Meek & Hildebrand, 1916; Reis *et al.*, 2003), whereas others have a wider range, such as *Sternopygus*, which is distributed up to the central Panama drainage of Santa María (Meek & Hildebrand, 1916), or *Gymnotus*, which holds two species: *G. henni*, found only in the Tuira basin (Alda *et al.*, 2013a), and *G. panamensis*, found in the Cricamola river on the western Atlantic coast (Albert & Crampton, 2003). *Brachyhypopomus occidentalis* (Regan, 1914) is the only gymnotiform that is widely distributed in Panama in nearly all Atlantic- and Pacific-slope drainages in all five biogeographical provinces (Eigenmann & Ward, 1905; Ellis, 1913), thus constituting an excellent study system in which to infer the patterns and processes of colonization across LCA. Its overall distribution extends from north-western Colombia and Venezuela to the Sixaola River in the Caribbean of Costa Rica (Meek & Hildebrand, 1916; Bussing, 1976; Reis *et al.*, 2003).

Previous studies of the diversification patterns of freshwater fishes in LCA showed that populations of *B. occidentalis* are well differentiated, and proposed an early colonization from SA around 7–4 Ma and subsequent isolation of the western and eastern populations around 3 Ma (Bermingham & Martin, 1998). Here, we use extensive geographical sampling and a multigene approach to construct a robust phylogeographical hypothesis about the tempo and mode of colonization of LCA by *B. occidentalis* and to understand the factors shaping its genetic structure, as well as to test specific biogeographical hypotheses based on the evolutionary relationships between biogeographical provinces.

MATERIALS AND METHODS

Sample collection

Specimens of *Brachyhypopomus occidentalis* were collected from 28 river drainages across Panama and some localities in north-western Colombia and Venezuela, using an electrofishing unit (see Appendix S1). We also used an electric-fish detector (Crampton *et al.*, 2007), consisting of two wire electrodes placed on a pole and connected to a mini amplifier–speaker (RadioShack, Fort Worth, TX, USA), which, when

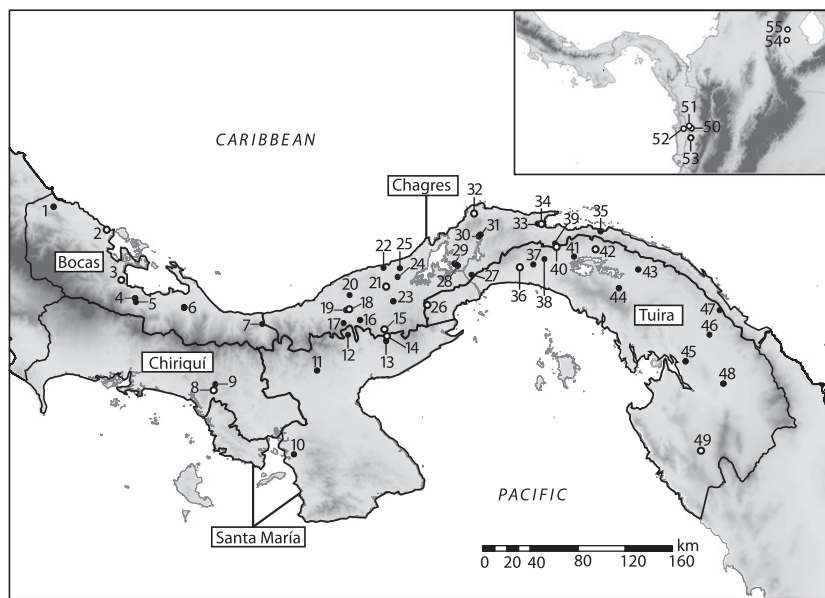


Figure 1 Map showing the localities of the samples of the weakly electric fish *Brachyhypopomus occidentalis* ($n = 109$) in lower Central America included in the analysis and the biogeographical regions defined by Smith & Bermingham (2005). The names of the biogeographical provinces are indicated in white rectangles. Black circles represent specimens used in the COI dataset and white circles are specimens used in the mtDNA and nDNA datasets. Numbers of localities correspond to those listed in Appendix S1.

submerged in the water, allows the localization of electric fish. Specimens were then collected using dip nets.

Fish were euthanized using an overdose of tricaine methanesulfonate (MS-222) (Ostrander, 2000). Gill arches with filaments were excised and preserved in saturated 20% dimethyl sulfoxide (DMSO) and 0.5 M EDTA pH 8 solution. Specimens were individually tagged and deposited in the Neotropical Fish Collection (NFC-STRI) at the Smithsonian Tropical Research Institute, Panama. All procedures were approved by the McGill University animal care committee.

DNA extraction, amplification, and sequencing

Our analysis includes five molecular markers: three mitochondrial genes (cytochrome *c* oxidase subunit I, *COI*; and ATP synthase 6 and 8, hereafter *ATPase 8/6*) and two nuclear genes (recombinant activating gene, *RAG1*; and rhodopsin). We sequenced the 5' region of the *COI* mtDNA gene (648 bp) for all collected *B. occidentalis* specimens ($n = 109$), representing 1–14 individuals per drainage. Additionally, the complete *ATPase 8/6* (842 bp) gene and partial *RAG1* (806 bp) and rhodopsin (780 bp) genes were sequenced for a reduced subset of 20 individuals representing all the evolutionary lineages identified in the *COI* dataset (Appendix S1; Fig. 1).

Total genomic DNA was isolated from gills using DNeasy Tissue Kits (Qiagen, Valencia, CA, USA) (see Appendix S2 for the primers and conditions used in PCR amplifications). Amplicons were sequenced in an ABI3130xl automated sequencer using the BigDye terminator v.3.1 kit (Applied Biosystems, Foster City, CA, USA).

Sequences were edited and visually aligned using GENEIOUS 5.6.3 (Drummond *et al.*, 2011). All sequences were translated into amino acids to check for stop codons, obvious sequencing errors or misalignments. No indels were found in any of the gene sequences.

Data analysis

Datasets and substitution models

Three datasets were constructed for phylogeographical analysis: one containing all 109 *B. occidentalis* individuals sequenced for *COI* (*COI* dataset); a second one containing the subset of 20 individuals sequenced for the mtDNA genes (mtDNA dataset); and a third one including the subset of 20 specimens sequenced for the nuclear genes (nDNA dataset). For all datasets, two accessions of *B. occidentalis* from the Maracaibo drainage of Venezuela were selected as geographical outgroups.

Prior to grouping different gene partitions together, the parsimony-based incongruence length difference test (ILD or partition homogeneity test in PAUP* 4.0b10) was performed to test the combinability of gene partitions (Swofford, 2002). Pairwise ILD comparisons indicated congruence between the three mitochondrial genes and between the two nuclear genes (mtDNA dataset, $P = 0.67$; nDNA dataset, $P = 0.51$).

The Akaike information criterion implemented in jMODELTEST 0.1.1 (Posada, 2008) was used to determine the evolutionary model that best fitted the data. Each gene was considered a different partition and the test was implemented separately for each of them. The models selected (Table 1) were used for Bayesian and maximum-likelihood phylogenetic analyses and divergence-time analyses.

Phylogeographical analyses

All datasets were subjected to Bayesian inference and maximum-likelihood methods. Bayesian inference (BI) was performed with MRBAYES 3.1.2 (Ronquist & Huelsenbeck, 2003): two simultaneous Markov chain Monte Carlo (MCMC) analyses were run for 5×10^6 generations, with

Table 1 Dimensions, polymorphism and substitution models selected by jMODELTEST for each of the datasets analysed

Dataset	Gene	<i>n</i>	Sequence length (bp)	Variable characters	Parsimony-informative characters	Substitution model
COI	COI	109	648	129 (19.91%)	118 (18.21%)	TIM1 + I + G
mtDNA	COI	20	648	112 (17.28%)	96 (14.81%)	TIM1 + I + G
	ATPase 8/6	20	842	164 (14.73%)	127 (15.08%)	GTR + G
nDNA	RAG1	20	806	27 (3.35%)	18 (2.23%)	TrNef + I
	Rhodopsin	20	780	18 (2.31%)	12 (1.54%)	HKY + I

trees sampled every 500 generations. Convergence between runs was assessed by monitoring the standard deviation of split frequencies in MRBAYES and using the effective sampling size (ESS) criterion in TRACER 1.5 (Rambaut & Drummond, 2007). By these measures, convergence was achieved within the first 25% of trees sampled, which were discarded as burn-in, and the remaining trees were taken as representative of the posterior probability distribution.

Maximum-likelihood (ML) analyses were performed using GARLI 2.0 (Zwickl, 2006). Five replicates of each search were run from random starting trees generated by a stepwise addition algorithm and 50 attachment attempts per taxon to find the tree with the highest likelihood. The maximum number of generations to be run to encounter a significantly better scoring topology (0.05) was set to 10,000. Support was estimated by 1000 bootstrap replicates.

For the COI dataset, uncorrected genetic pairwise (*p*)-distances between individuals, as well as mean *p*-distances between and within clades were calculated using MEGA 5 (Tamura *et al.*, 2011). Mean between- and within-clade distances were also calculated for the mtDNA and nDNA datasets. Standard errors (SE) were estimated by bootstrapping (1000 replicates) in MEGA.

Time of divergence estimates

The time to the most recent common ancestor and its confidence interval (95% highest posterior density: HPD) was estimated for each clade of the mtDNA dataset, using a relaxed molecular clock with uncorrelated lognormal distribution of rates in BEAST 1.6.2 (Drummond & Rambaut, 2007). A Yule speciation model was assumed as a tree prior, i.e. a constant speciation rate per lineage (Drummond *et al.*, 2006). Four independent analyses were performed, running the MCMC for 4×10^7 generations, with trees sampled every 4000 generations. Runs were checked for convergence and adequate ESS (all ESS ≥ 8000) using TRACER, and combined using LOGCOMBINER 1.6.2 with a burn-in fraction of 20%. The final tree was produced using TREEANNOTATOR 1.6.2 (available at <http://beast.bio.ed.ac.uk/TreeAnnotator>).

To calibrate the molecular clock, we used two calibration points. First, we used a fossil of *Humboldtichthys kirschbaumi* dated to c. 10 Ma, which shares morphological characters of the opercle with extant *Sternopygus* species (Albert & Fink, 2007). We included additional sequences from four genera of gymnotiforms – *Apteronotus*, *Eigenmannia*, *Gymnotus* and

Sternopygus – and placed a minimum age constraint of 10 Ma on the stem node of the genus *Sternopygus*, using a uniform prior distribution with a maximum age equal to the maximum tree root height. Second, we used the uplift of the Andes to calibrate the divergence of cis-Andean *Brachyhypopomus* in Venezuela from trans-Andean populations in Colombia and Panama. Previous (Bermingham & Martin, 1998) and current phylogenetic analyses show that these groups are sister clades and, as a result, we assume that their divergence pre-dates the uplift of the Eastern Cordillera of the Andes, whose highest peak of tectonic activity occurred around 12.9 Ma (Lundberg *et al.*, 1998; Albert *et al.*, 2006; Hoorn *et al.*, 2010). Thus, we placed a normal prior with a mean of 12.9 Ma and a wide confidence interval (standard deviation = 2 Myr) at the crown node of the *Brachyhypopomus* clade.

Test of biogeographical hypotheses

Because the embryonic Isthmus might have provided different opportunities for colonization, we proposed several simplified biogeographical scenarios to test which of them is fitted best by our data. We constructed three area cladograms (scenarios 1, 2 and 3) by replacing individual taxa from the mtDNA dataset with the biogeographical province (defined by Smith & Bermingham, 2005; Fig. 1) that corresponds to their location. The scenarios are based on different colonization patterns of LCA as previously proposed for freshwater fishes.

Scenario 1 assumes a single stepwise colonization from south to north (e.g. *Poecilia sphenops* complex, Alda *et al.*, 2013b; *Synbranchus* and *Ophisternon* eels, Perdices *et al.*, 2005; and *Cyphocharax*, Reeves & Bermingham, 2006), originating in SA and leading to the establishment of populations first in north-western Colombia, and progressively in the Tuirá, Chagres, Santa María, Chiriquí and Bocas provinces (Fig. 2a). Scenario 2 assumes two independent colonization events, both originating in SA [e.g. *Rhamdia* (Perdices *et al.*, 2002), *Pimelodella*, *Roeboidea*, *Brycon* and *Bryconamericus* (Bermingham & Martin, 1998; Reeves & Bermingham, 2006), Heroini cichlids (Říčan *et al.*, 2013)]: the first one leading to establishment only in the Bocas province, and the second occurring again as a stepwise colonization from south to north, first establishing in north-western Colombia and then progressing into the Tuirá, Chagres, Santa María and Chiriquí provinces (Fig. 2b). Scenario 3 is identical to Scenario 2, except that the three Pacific biogeographical provinces of

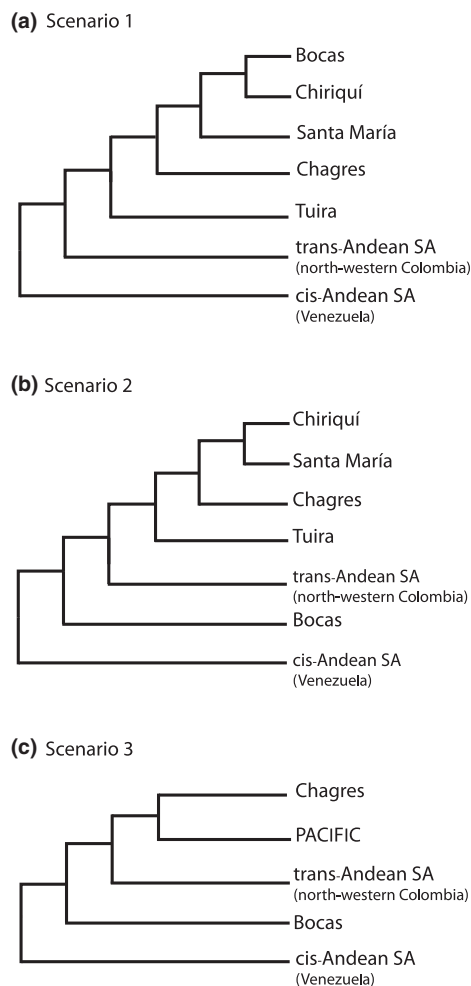


Figure 2 Area cladograms representing three biogeographical scenarios tested to characterize the colonization pattern of the weakly electric fish *Brachyhypopomus occidentalis* over the lower Central American landscape. (a) Scenario 1 assumes a single stepwise colonization from south to north; (b) Scenario 2 assumes two independent colonization events; and (c) Scenario 3 also assumes two independent colonization events but groups the three Pacific biogeographical provinces of Tuira, Santa María and Chiriquí into a single region (PACIFIC). Areas represent the biogeographical provinces defined by Smith & Bermingham (2005) (see Fig. 1).

Tuira, Santa María and Chiriquí are considered as a single province referred to as PACIFIC, to account for the dispersal pathway proposed to have existed along the Pacific coast during marine regressions in glacial maxima (Loftin, 1965; Bermingham & Martin, 1998; Smith & Bermingham, 2005) (Fig. 2c).

We used Bayes factors (BF) to test which biogeographical scenario fitted our data best. For each scenario, we ran BEAST 1.6.2, fixing the tree topology to the corresponding area cladogram, keeping all tree priors as in the molecular clock analyses, except that the MCMC analyses were run for 5×10^7 generations with trees sampled every 5000 generations. The unconstrained mtDNA dataset was also run under

the same parameters. We then calculated BF in TRACER, by comparing twice the difference in the marginal model posterior likelihoods as estimated from the harmonic mean of the sample of posterior trees for each of the pairs of the proposed and unconstrained biogeographical scenarios. Following the guidelines of Kass & Raftery (1995), values of $2 \times \log_{10}(\text{BF}) > 20$ were considered to be strong evidence for a given model to be more likely than another, and values of $2 \times \log_{10}(\text{BF}) > 200$ as decisive evidence.

RESULTS

Phylogeographical relationships

All sequences obtained have been deposited in GenBank (KJ000886–KJ000994; KJ135099–KJ135119) (see Table 1 for the numbers of samples, sequence lengths and polymorphisms of each dataset and gene partition).

Independent BI and ML analyses of the *COI* dataset produced congruent phylogeographical hypotheses (Fig. 3). Both hypotheses support the monophyly of trans-Andean *B. occidentalis* (Panama and Western Colombia) and show high phylogeographical structure, allowing two major lineages to be identified: the Bocas lineage (posterior probability, PP, 1; bootstrap value, BS, 100%) and the Panama–Colombia lineage (PP, 1; BS, 66%) (Fig. 3), diverging by a mean *COI* *p*-distance of 5.9% (SE = 0.7%; Table 2). The Bocas lineage contains all individuals from the westernmost Atlantic drainages of Sixaola, Changuinola, Cricamola and Calovébora ($n = 20$) and the Panama–Colombia lineage contains all the remaining individuals from Panama and Colombia ($n = 87$). Within the latter lineage, four main clades with clear geographical distributions were recovered with high posterior probability and bootstrap support (PP, 1; BS $\geq 72\%$). From west to east, we identified and named the following clades according to the main drainage basins where they occur: Coclé del Norte (CN); Chagres (CHA); BAHIA, which holds the Pacific basins of Chiriquí–Santa María, Bayano and Tuira; and the cross-Cordillera clade (CC), which is the only clade with individuals from both sides of the Central Cordillera (Pacora and Bayano in the Pacific side and San Blas in the Atlantic), and the Atrato, Baudó and San Juan basins in Colombia (Fig. 3).

The highest mean genetic distance within the Panama–Colombia lineage was found between CHA and CC (5.5%, SE = 0.7%), and the lowest was between BAHIA and CN (3.9%, SE = 0.6%). Within clades, Bocas (0.22%, SE = 0.08%) and CC (1.6%, SE = 0.3%) showed the smallest and largest distances, respectively. Sequence divergence was consistently low among specimens from the same drainage. One exception was the Bayano drainage, where specimens from the upper streams were found in the BAHIA clade and diverged by more than 2% from the lower streams, which were found in the CC clade (Fig. 3).

The analysis of the complete mtDNA dataset (*COI* and *ATPase 8/6*) yielded topologies that were mostly congruent

with the *COI* gene tree, with the exception of some unresolved relationships between the CHA, CN and BAHIA clades in the Panama–Colombia lineage (compare Figs 3 & 4a). On the other hand, the hypothesis based on nDNA data only recovered the Bocas lineage with high support (PP, 1; BS > 90%), and showed a mean *p*-distance of 0.6% (SE = 0.2%) from the remaining individuals with a Panama–Colombia distribution. The relationships between individuals from the Panama–Colombia lineage remained unresolved and no geographical structure was observed (Fig. 4b).

Divergence time estimates

The molecular-clock analysis indicated that the time of divergence between the Bocas and Panama–Colombia lineage was in the late Miocene, around 7.77 Ma (95% HPD: 11.03–4.44 Ma) (point 1 in Fig. 5). Subsequently, the major diversification events generating the four clades of the Panama–Colombia lineage were estimated to occur in the early Pliocene, about 5.38 Ma (95% HPD: 7.72–3.14 Ma) (point 2 in Fig. 5). Finally, the most recent divergence event, between populations from north-western Colombia and Panama (PAC–BAY–SB), was estimated to occur in the late Pliocene, around 2.79 Ma (95% HPD: 4.36–1.42) (point 3 in Fig. 5), indicating very recent contact between these populations. Population differentiation was estimated to be younger than 2.8 Ma.

The mean substitution rates were estimated as 0.00675 substitutions site⁻¹ Myr⁻¹ (95% HPD: 0.00100–0.00403 s s⁻¹ Myr⁻¹) for *ATPase 8/6* and 0.00545 s s⁻¹ Myr⁻¹ (0.00332–0.00826 s s⁻¹ Myr⁻¹) for *COI*. Both rates fall well within the range of those previously reported for other freshwater fishes, ranging from 0.0045 to 0.006 s s⁻¹ Myr⁻¹ for *COI* and 0.0054–0.0066 s s⁻¹ Myr⁻¹ for *ATPase 8/6* (Bermingham *et al.*, 1997; Machordom & Doadrio, 2001; Sivasundar *et al.*, 2001; Webb *et al.*, 2004; reviewed in Burrridge *et al.*, 2008).

Test of biogeographical hypotheses

Among the proposed biogeographical scenarios, Scenario 3 showed the best fit ($2 \times \log_{10}(\text{BF}) = 152.19$ and 28.42, when compared with scenarios 1 and 2, respectively), which supports the existence of multiple colonization events with a single Pacific biogeographical province. Scenario 2 was more strongly supported than Scenario 1 ($2 \times \log_{10}(\text{BF}) = 123.77$), emphasizing that a model assuming a single colonization event was the least likely for *B. occidentalis*.

DISCUSSION

The current study represents one of the most detailed phylogeographical analyses so far of a primary freshwater fish in LCA. We provide evidence that multiple lineages may have colonized LCA independently. Early colonization of this region by *B. occidentalis* occurred during the Miocene. Only

one spatially restricted lineage remains from this event, which was followed by more extensive colonization during the Pliocene, probably facilitated by the completion of the Isthmus of Panama. During this process, the history of formation of the embryonic Isthmus had a profound effect on the distributions of fish lineages.

Geographical partitioning of genetic variation

Brachyhypopomus occidentalis showed strong phylogeographical structure across the Isthmian landscape, characterized by clear breaks between geographically non-overlapping clades. Lineages and clades were not structured according to a distinct directional pattern, nor according to a clear division of Atlantic- and Pacific-slope drainages, and closely related subclades did not necessarily represent geographical neighbours. This suggests that the phylogeographical structure reflects the progressive complex and dynamic geological evolution of the area.

Deeper genetic divergences were revealed over the Caribbean slope than over the Pacific slope. The specimens from the four major Atlantic drainages belonged to four separate and well-supported clades: Bocas, CN, CHA and Pac-Bay-SB (Figs 3 & 4), confirming that Atlantic drainages appear to be historically more isolated (Bermingham & Martin, 1998). Conversely, the Pacific slope showed reduced genetic structure, in that all drainages (except for Pacora) grouped into the BAHIA clade (Figs 3 & 4), forming three subclades joined by short internodes, suggesting rapid dispersal and expansion events across the Pacific slope. This Pacific dispersal corridor was also supported by our biogeographical hypothesis tests, which showed that the scenarios considering the three Pacific biogeographical provinces of Chiriquí, Santa María and Tuira as a single province were best fitted by our data, confirming a shared history for *B. occidentalis* in these drainages. This pattern can be explained by events such as Pleistocene glaciations, which exposed the low gradient of the Pacific continental shelf above sea levels and permitted the emergence of lowland rivers and swamps, facilitating the dispersal of freshwater fishes (Loftin, 1965; Bermingham & Martin, 1998).

It is noteworthy that all the phylogenetic breaks in *B. occidentalis* coincided with the Central Cordillera, confirming its role as a strong barrier to dispersal, except in two cases (CHA and CC clades; Figs 3 & 4). For the CHA clade, dispersal and a lack of genetic structure across both slopes is unsurprising for two reasons. First, in Central Panama, the cordillera is characterized by very low elevations; second, the Panama Canal, by breaching the continental divide, has created a direct freshwater connection between the Chagres and Grande rivers and dramatically increased the rate of cross-Cordillera exchanges (Smith *et al.*, 2004). In the case of the CC clade, however, the genetic similarity between the drainages of San Blas (Chagres province) and Pacora and Bayano (Tuirá province) is remarkable, because the high elevation of the Cordillera in that region is expected to act as an impor-

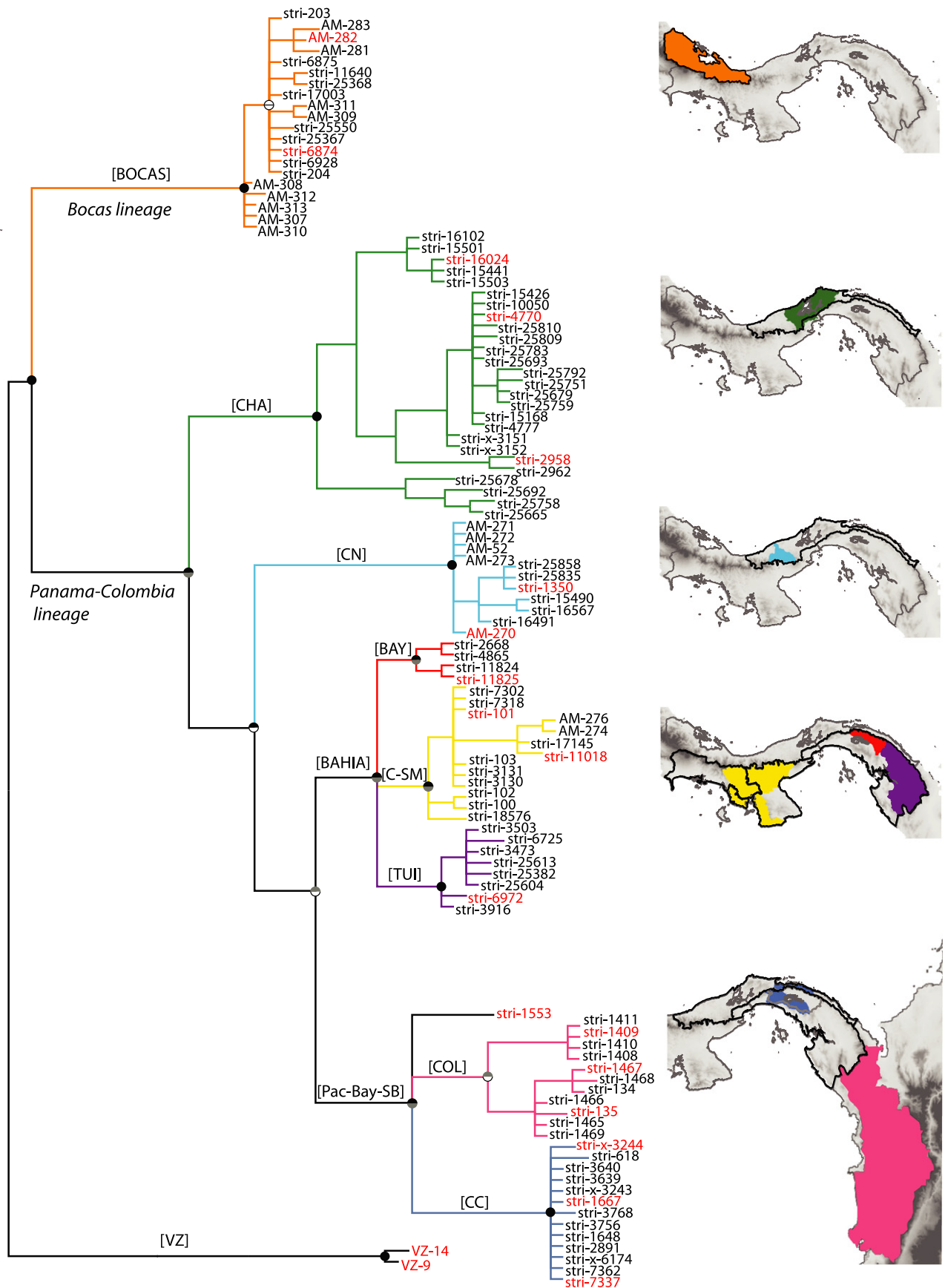


Table 2 Mean uncorrected *p*-distances for the *COI* gene between the main clades of *Brachyhypopomus occidentalis*. Numbers in parentheses are standard errors. Numbers in italics in the diagonal are the mean within-clade uncorrected *p*-distances for the *COI* gene.

	Bocas	Panama–Colombia lineage			
		CN	CHA	BAHIA	CC
Bocas	<i>0.0022 (0.0008)</i>				
CN	0.0603 (0.0093)	<i>0.0025 (0.0012)</i>			
CHA	0.0570 (0.0085)	0.0450 (0.0067)	<i>0.0133 (0.0023)</i>		
BAHIA	0.0565 (0.0082)	0.0393 (0.0061)	0.0399 (0.0060)	<i>0.0106 (0.0023)</i>	
CC	0.0636 (0.0088)	0.0490 (0.0068)	0.0555 (0.0073)	0.0406 (0.0062)	<i>0.0161 (0.0031)</i>

tant geophysical barrier to dispersal. The low genetic distance suggests recent contact between Atlantic and Pacific drainages, and a fairly contemporaneous cross-Cordillera dispersal event. This could have been facilitated by head-river captures between the lower rivers of Bayano and the Atlantic San Blas drainage, with the upper rivers of Bayano included in the BAHIA clade together with the other Pacific drainages, probably because of a past connection with the Tuira River (Figs 3 & 4). This could also have occurred via geodispersal (Albert & Carvalho, 2011), which would imply partial erosion of the Cordillera barrier allowing for transient contact between these populations.

Given the division of biogeographical provinces along the Central Cordillera (Smith & Bermingham, 2005), the cross-Cordillera exchange of *B. occidentalis* appears to be a rather exceptional, although not unique, pattern for freshwater fish. Similar crossings of geographical boundaries and biogeographical provinces have been observed in a few other species (e.g. *Brycon*; Reeves & Bermingham, 2006). Further work is required to identify the biotic or abiotic processes that lead to this phylogeographical pattern.

The large phylogenetic gap separating the Bocas and Panama–Colombia lineages (Fig. 4a) despite a lack of obvious physiographical barriers is a striking result. This split was the only one supported by the nuclear phylogeny, whereas the other drainages of Panama showed no genetic structure according to their nuclear genes (Fig. 4b). Similar discontinuities found in the distributions of other freshwater fishes have been attributed to the high gradient of the continental shelf in north-western Panama, which may have reduced the probability of river anastomosis (Bussing, 1985; Smith & Bermingham, 2005). Interestingly, several other studies have detected deep genetic breaks in this region, referred to as the ‘Bocas break’ (Crawford *et al.*, 2007), notably in primary

freshwater fishes (Perdices *et al.*, 2002; Reeves & Bermingham, 2006), frogs (Crawford, 2003; Crawford *et al.*, 2007; Wang *et al.*, 2008), caimans (Venegas-Anaya *et al.*, 2008), mosquitoes (Loaiza *et al.*, 2010) and trees (Dick *et al.*, 2003), although with different divergence rates and presumably therefore different isolation times and processes.

Timing of colonization and diversification

Biogeographical analyses demonstrated stronger support for two colonizations than for a single colonization by *B. occidentalis*, with the Bocas populations being established first. We conclude that *B. occidentalis* first colonized CA in the late Miocene, with the westernmost Atlantic populations of Bocas as the only relicts of this earliest wave of colonization. The high genetic divergence between Bocas and Panama–Colombia and the remarkably long branch of the latter between its stem node at 7.72 Ma and its first recorded diversification at 5.38 Ma (Fig. 5), suggest a long-term stasis in diversification or large extinction events for populations in central and eastern Panama and subsequent colonization from SA. The limited distribution of the Bocas lineage and the relative delay before the Panama–Colombia lineage recolonized and became established in central–eastern Panama suggest that the LCA landscape was not equally conducive to dispersal throughout its development; therefore, early colonizations during the rise of the Isthmus are likely to represent restricted events that did not lead to the successful establishment of freshwater fish lineages throughout the region.

Which geological hypotheses could explain an early dispersal event? Neither Bussing’s proposed Cretaceous proto-Antillean connection between SA and CA c. 65 Ma (1976, 1985) nor the GAARlandia hypothesis, which proposed a land connection between northern CA and north-western SA

Figure 3 Phylogenetic tree rendered by Bayesian inference analysis of the mitochondrial *COI* gene for 109 *Brachyhypopomus occidentalis* samples from lower Central America. Bullets on nodes indicate posterior probabilities (upper half) and bootstrap values (lower half) for the maximum likelihood analysis. Black denotes highly-supported nodes (PP ≥ 0.99; BS ≥ 90%), grey denotes well-supported nodes (PP ≥ 0.90; BS ≥ 65%), and white denotes low (PP < 0.90; BS < 65%) or no support. Sample IDs as indicated in Appendix S1. Sample IDs in red represent the 20 individuals included in the mtDNA and nDNA datasets. Letters in italics below branches indicate lineage names, and names in square brackets above branches indicate clade names: Bocas, Chagres (CHA), Coclé del Norte (CN), Pacific Bay of Panama (BAHIA), Chiriquí–Santa María (C-SM), Bayano (BAY), Tuira (TUI), cross-Cordillera (CC), Pacora–Bayano–San Blas (Pac-Bay-SB), Colombia (COL) and Venezuela (VZ). Maps to the right schematically represent the distribution of each identified clade. The biogeographical regions defined by Smith & Bermingham (2005) are outlined in black in the maps.

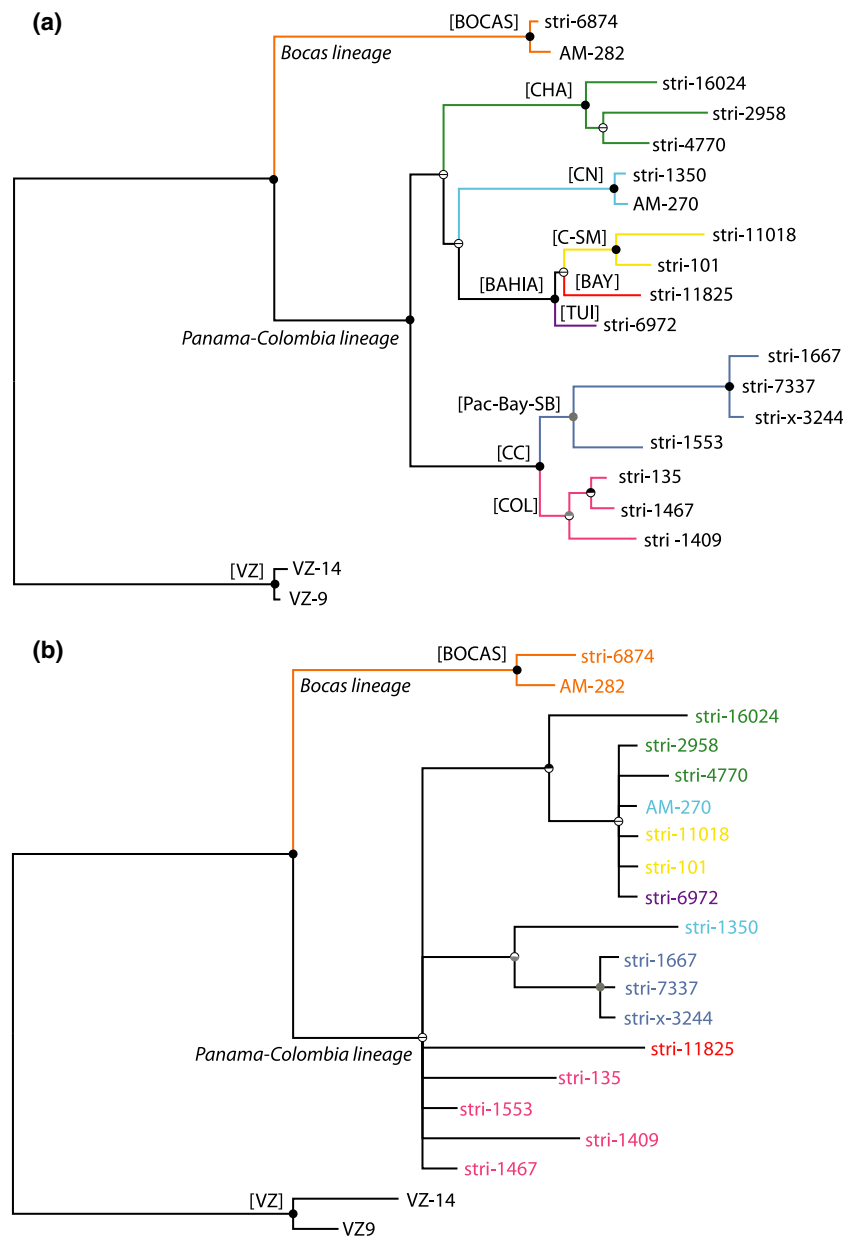


Figure 4 Phylogenetic trees rendered by Bayesian inference analysis of (a) mitochondrial genes (*COI* and *ATPase 8/6*) and (b) nuclear genes (*RAG1* and rhodopsin) for 20 *Brachyhytopomus occidentalis* samples from lower Central America. Bullets on nodes indicate posterior probabilities (upper half) and bootstrap values (lower half) for the maximum likelihood analysis. Black denotes highly-supported nodes (PP ≥ 0.99 ; BS ≥ 90), grey denotes well-supported nodes (PP ≥ 0.90 ; BS ≥ 65), and white denotes low (PP < 0.90 ; BS < 65) or no support. Sample IDs as indicated in Appendix S1. Colours refer to the distribution of identified clades shown on the maps in Fig. 3. Letters in italics below branches indicate lineage names, and names between brackets above branches indicate clade names: Bocas, Chagres (CHA), Coclé del Norte (CN), Pacific Bay of Panama (BAHIA), Chiriquí-Santa María (C-SM), Bayano (BAY), Tuira (TUI), cross-Cordillera (CC), Pacora-Bayano-San Blas (Pac-Bay-SB), Colombia (COL), and Venezuela (VZ).

through the Greater Antilles 35–33 Ma (Iturralde-Vinent & MacPhee, 1999), matches our estimated dates, in that they pre-date by more than 50 and 20 million years, respectively, our upper 95% HPD estimates for the first colonization of CA by *B. occidentalis*. On the other hand, our estimates for the first colonization event are close to the most recently hypothesized final closure of the Isthmus at *c.* 15 Ma (Montes *et al.*, 2012a,b), which implies that a continuous

terrestrial corridor has existed since the late Miocene. The observed lag in the diversification of *B. occidentalis* in central and eastern drainages, however, does not agree with this geological model, and further vicariance events need to be invoked in order to explain the great divergence and allopatric distribution of lineages within *B. occidentalis*. For example, the gradual uplift of the Isthmus (as described by Coates & Obando, 1996), combined with significant sea-level

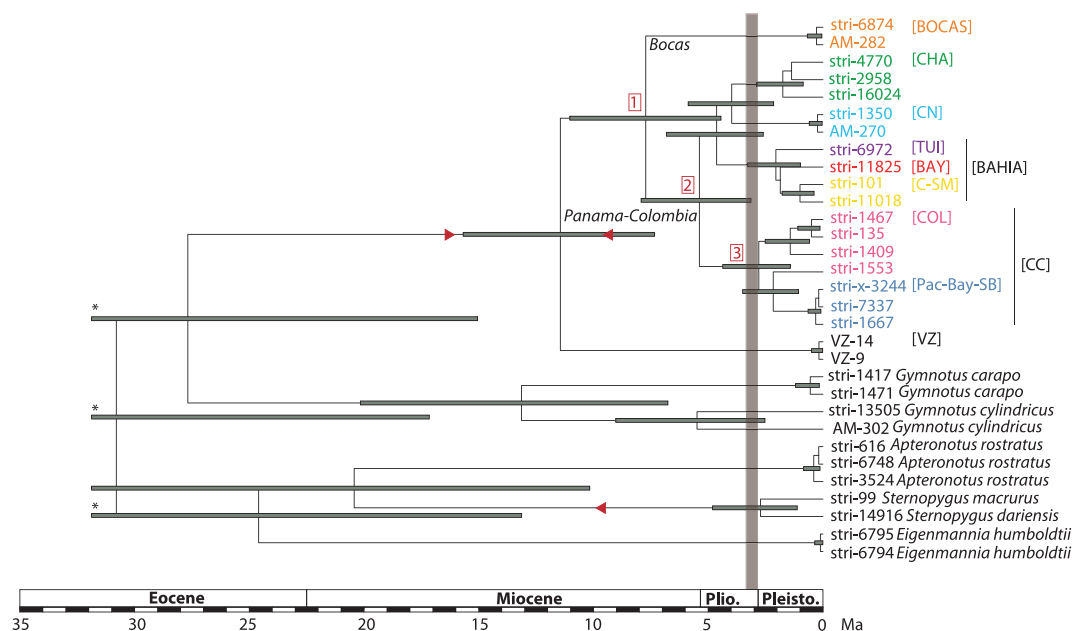


Figure 5 Chronogram of the lower Central American weakly electric fish *Brachyhypopomus occidentalis* derived from a relaxed-clock Bayesian analysis based on the mtDNA dataset. Horizontal grey bars are 95% highest probability density (HPD) for age estimates using calibration rates based on fossil data and the uplift of the eastern cordillera of the Andes (see Materials and Methods). 95% HPD bars were only drawn for nodes with posterior probability > 0.75. Sample IDs are detailed in Appendix S1. Names between square brackets indicate clade names: Bocas, Chagres (CHA), Coclé del Norte (CN), Pacific Bay of Panama (BAHIA), Chiriquí-Santa María (C-SM), Bayano (BAY), Tuira (TUI), cross-Cordillera (CC), Pacora-Bayano-San Blas (Pac-Bay-SB), Colombia (COL) and Venezuela (VZ). Colours refer to the distributions of the identified clades shown on the maps in Fig. 3. Asterisks indicate that the confidence interval bars are truncated at that end. Divergence events discussed in the text are indicated on the relevant nodes of the tree with red numbers. Minimum and maximum age constraints used for calibration are represented by red arrows to the left and right, respectively (see Materials and Methods). The time chart shows the duration of the following epochs: Pleistocene (Pleisto.), Pliocene (Plio.), Miocene and Eocene. The vertical grey bar spanning 3.5–3 Ma represents the closure of the Isthmus of Panama as defined by Coates *et al.* (1992, 2004) and Coates & Obando (1996).

regressions in the late Miocene, could have resulted in a short-lived emergence of land, providing a first opportunity for the colonization of LCA by *B. occidentalis* (Bermingham & Martin, 1998). Subsequently, major inundations of central and eastern Panama through land submergence or sea-level transgressions during interglacial periods (Haq *et al.*, 1987) may have led to regional extinctions, leaving a gap in the distribution of *B. occidentalis* in Panama. The closely related genus *Gymnotus* presents a similar pattern of a Miocene colonization of CA estimated at c. 15 Ma (95% HPD: 7.18–24.28) (Lovejoy *et al.*, 2010). It is currently represented by several lineages – or species – from Bocas up to Mexico, and a probably more recent colonization restricted to the Tuira drainage (Alda *et al.*, 2013a), leaving a distribution gap in the central region of the Isthmus of Panama. Therefore, integrating seawater regression and transgression cycles into the new geological model of Isthmus uplift (Farris *et al.*, 2011; Montes *et al.*, 2012a,b) could help us to understand how this corridor contributed to primary freshwater fish dispersal.

An alternative scenario congruent with our time estimates is that early colonization of CA was facilitated by the strong northward discharge of the palaeo-Amazon–Orinoco in the region of the Magdalena and Maracaibo 8.5–8 Ma (Lundberg *et al.*, 1998) when LCA may have been an island landscape

relatively close to SA, only separated by shallow seawater connections between the Caribbean and Pacific oceans (Coates *et al.*, 1992, 2004; Coates & Obando, 1996; but see Kirby & MacFadden, 2005).

Following the Miocene colonization, we propose a second major colonization of central and eastern Panama 7.72–3.14 Ma. The 95% HPD estimates for this dispersal agree with the standard palaeogeographical reconstructions of the Isthmus emerging from the early Pliocene until its final closure around 3.5–3 Ma (Coates *et al.*, 1992, 2004; Coates & Obando, 1996). The short internodes connecting the mtDNA clades composing the Panama–Colombia lineage suggest a rapid expansion event, probably allowing for secondary connections between LCA and northern SA rivers. The wide geographical distribution of these mtDNA clades over the Isthmus also suggests that the LCA landscape was then already acting as a strong dispersal corridor, a pattern that agrees with its colonization by many other fish groups (Bermingham & Martin, 1998; Martin & Bermingham, 2000; Perdices *et al.*, 2002; Reeves & Bermingham, 2006).

In conclusion, our study suggests a complex biogeographical history of multiple colonization, diversification and extinction events over the Isthmian landscape, demonstrating the key role played by the complex geological evolution of

the region in the distribution of freshwater fishes. Our phylogeographical hypothesis showed high structure, with most lineages colonizing CA in the middle–late Pliocene, when the Isthmus already constituted a continuous land bridge. Nonetheless, we demonstrate that faunal connections were also possible between CA and SA before the late Pliocene, although these may have been ‘sporadic’ events for some taxa. The mode of these early colonizations and the factors limiting them remain to be fully understood.

ACKNOWLEDGEMENTS

We wish to thank all the people who helped in the collection of specimens, especially Andrew Martin, Rigoberto González and Ruth G. Reina. We are also very grateful to Rigoberto González for curatorial assistance and to Ruth G. Reina for laboratory assistance. We thank Vielka Salazar for insightful comments while preparing this manuscript as well as two anonymous reviewers and the editor for helpful suggestions that significantly improved this manuscript. Financial and logistical support was provided by grants from the McGill-Smithsonian Tropical Research Institute Neotropical Environment Program to S.P., from NSERC (Natural Sciences and Engineering Research Council of Canada) to R.K., and from the Smithsonian Institution to E.B.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Appendix S1 Sampling table and GenBank accession numbers.

Appendix S2 Primers and PCR conditions used for amplification and sequencing.

BIOSKETCH

Sophie Picq recently completed her MSc in Biology at McGill University in the Neotropical Environment Option (NEO) program. She is interested in the evolutionary and ecological processes that shape and maintain biological diversity, mostly in tropical freshwater ecosystems.

The main research interests of the authors are the evolution, phylogenetics, and historical biogeography of Neotropical fishes.

Author contributions: This study was designed jointly by all authors and formed part of S.P.'s MSc thesis, supervised by E.B. and R.K.; S.P. conducted the molecular laboratory work and, together with F.A., performed the phylogenetic and biogeographical analyses; E.B. provided laboratory space, as well as financial and technical support for the collection of specimens and molecular laboratory work; S.P. led the writing and all authors discussed the results and commented on the manuscript.

Editor: Malte Ebach